

Homework 3

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1 Problem 1

The FTCS discretization of the diffusion equation is

$$T_j^{n+1} = T_j^n + \frac{\kappa \Delta t}{(\Delta x)^2} (T_{j+1}^n - 2T_j^n + T_{j-1}^n).$$

It is useful to define

$$\alpha \equiv \frac{\kappa \Delta t}{(\Delta x)^2},$$

so that

$$T_j^{n+1} = \alpha T_{j-1}^n + (1 - 2\alpha)T_j^n + \alpha T_{j+1}^n.$$

(a)

Writing the update as

$$\mathbf{T}^{n+1} = A\mathbf{T}^n,$$

one row of the matrix A , away from the boundaries, is

$$A_{j,:} = \begin{pmatrix} \cdots & \alpha & 1 - 2\alpha & \alpha & \cdots \end{pmatrix}.$$

Thus, the three nonzero elements in an interior row are

$$A_{j,j-1} = \alpha, \quad A_{j,j} = 1 - 2\alpha, \quad A_{j,j+1} = \alpha.$$

(b)

For the von Neumann stability analysis, assume a Fourier mode of the form

$$T_j^n = \xi^n(k) e^{ikj\Delta x}.$$

Substituting this into the FTCS update gives

$$\xi^{n+1} e^{ikj\Delta x} = \xi^n e^{ikj\Delta x} + \alpha \left[\xi^n e^{ik(j+1)\Delta x} - 2\xi^n e^{ikj\Delta x} + \xi^n e^{ik(j-1)\Delta x} \right].$$

Dividing by $\xi^n e^{ikj\Delta x}$ gives

$$\xi = 1 + \alpha \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right).$$

Using

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

this becomes

$$\xi = 1 + 2\alpha (\cos(k\Delta x) - 1).$$

The identity

$$\cos \theta - 1 = -2 \sin^2 \left(\frac{\theta}{2} \right)$$

then gives

$$\boxed{\xi(k) = 1 - 4 \frac{\kappa \Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k \Delta x}{2} \right)}.$$

(c)

For the method to be stable, every Fourier mode must satisfy

$$|\xi(k)| \leq 1.$$

Since $\alpha > 0$, the largest possible value of ξ is already 1. The lower bound therefore gives the timestep restriction,

$$1 - 4\alpha \sin^2\left(\frac{k\Delta x}{2}\right) \geq -1.$$

The most restrictive mode occurs when

$$\sin^2\left(\frac{k\Delta x}{2}\right) = 1.$$

Therefore,

$$1 - 4\alpha \geq -1,$$

so that

$$4\alpha \leq 2,$$

or

$$\alpha \leq \frac{1}{2}.$$

Substituting the definition of α ,

$$\frac{\kappa\Delta t}{(\Delta x)^2} \leq \frac{1}{2},$$

which gives

$$\boxed{\Delta t \leq \frac{(\Delta x)^2}{2\kappa}}.$$

(d)

For a fixed physical domain,

$$\Delta x = \frac{x_{\max} - x_{\min}}{J}.$$

Doubling the spatial resolution means replacing J by $2J$, so

$$\Delta x \rightarrow \frac{\Delta x}{2}.$$

Since the stable timestep scales as

$$\Delta t_{\max} \propto (\Delta x)^2,$$

the new timestep must satisfy

$$\Delta t_{\max} \rightarrow \frac{1}{4} \Delta t_{\max}.$$

Therefore, doubling the spatial resolution requires reducing the timestep by a factor of

$$\boxed{4}.$$

2 Problem 2

The fully implicit scheme evaluates the spatial derivative using the new temperature values at time $n + 1$,

$$T_j^{n+1} = T_j^n + \alpha \left(T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} \right),$$

where again

$$\alpha = \frac{\kappa \Delta t}{(\Delta x)^2}.$$

(a)

Collecting all of the $n + 1$ terms on the left gives

$$-\alpha T_{j-1}^{n+1} + (1 + 2\alpha) T_j^{n+1} - \alpha T_{j+1}^{n+1} = T_j^n.$$

For boundary conditions

$$T_0 = T_J = 0,$$

only the $J - 1$ interior temperatures need to be solved for. Defining

$$\mathbf{T}_{\text{int}}^n = \begin{pmatrix} T_1^n \\ T_2^n \\ \vdots \\ T_{J-2}^n \\ T_{J-1}^n \end{pmatrix},$$

the update can be written as

$$B\mathbf{T}_{\text{int}}^{n+1} = \mathbf{T}_{\text{int}}^n,$$

with the tridiagonal matrix

$$B = \begin{pmatrix} 1+2\alpha & -\alpha & 0 & \cdots & 0 \\ -\alpha & 1+2\alpha & -\alpha & \ddots & \vdots \\ 0 & -\alpha & 1+2\alpha & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\alpha \\ 0 & \cdots & 0 & -\alpha & 1+2\alpha \end{pmatrix}.$$

The boundary rows do not need to be included because the two wall temperatures remain fixed at zero.

(b)

Using the Fourier mode

$$T_j^n = \xi^n e^{ikj\Delta x},$$

the implicit update becomes

$$\xi^{n+1} e^{ikj\Delta x} = \xi^n e^{ikj\Delta x} + \alpha \left[\xi^{n+1} e^{ik(j+1)\Delta x} - 2\xi^{n+1} e^{ikj\Delta x} + \xi^{n+1} e^{ik(j-1)\Delta x} \right].$$

Dividing by $\xi^n e^{ikj\Delta x}$ gives

$$\xi = 1 + \alpha\xi \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right).$$

As in Problem 1,

$$e^{ik\Delta x} - 2 + e^{-ik\Delta x} = -4 \sin^2 \left(\frac{k\Delta x}{2} \right),$$

so

$$\xi = 1 - 4\alpha\xi \sin^2 \left(\frac{k\Delta x}{2} \right).$$

Collecting the factors of ξ ,

$$\xi \left[1 + 4\alpha \sin^2 \left(\frac{k\Delta x}{2} \right) \right] = 1.$$

Therefore,

$$\xi(k) = \frac{1}{1 + 4\frac{\kappa\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right)}.$$

(c)

For

$$\kappa > 0, \quad \Delta t > 0,$$

the denominator of ξ is always greater than or equal to one. Therefore,

$$0 < \xi(k) \leq 1$$

for every wavenumber and every positive timestep.

Thus the fully implicit method is unconditionally stable and there is no finite timestep restriction from stability,

$$\Delta t \text{ is unrestricted by stability.}$$

Very large timesteps may still give a poor approximation to the time evolution, but they do not make the numerical solution unstable.

(d)

Taking the limit

$$\Delta t \rightarrow \infty$$

also gives

$$\alpha \rightarrow \infty.$$

For every nonconstant Fourier mode,

$$\xi(k) = \frac{1}{1 + 4\alpha \sin^2(k\Delta x/2)} \rightarrow 0.$$

Since the temperatures at both boundaries are fixed to zero, the equilibrium solution of the diffusion equation is

$$T(x) = 0.$$

Therefore, in the large-timestep limit,

$$\boxed{T_j \rightarrow 0}$$

at every interior grid point.

3 Problem 3

The Crank–Nicholson method averages the explicit and implicit spatial derivatives. Its update rule is therefore

$$\frac{T_j^{n+1} - T_j^n}{\Delta t} = \frac{\kappa}{2(\Delta x)^2} \left[T_{j+1}^n - 2T_j^n + T_{j-1}^n + T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} \right].$$

Using

$$\alpha = \frac{\kappa \Delta t}{(\Delta x)^2},$$

this can also be written as

$$T_j^{n+1} = T_j^n + \frac{\alpha}{2} \left[T_{j+1}^n - 2T_j^n + T_{j-1}^n + T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} \right].$$

Collecting the new-time terms on the left gives

$$-\frac{\alpha}{2} T_{j-1}^{n+1} + (1 + \alpha) T_j^{n+1} - \frac{\alpha}{2} T_{j+1}^{n+1} = \frac{\alpha}{2} T_{j-1}^n + (1 - \alpha) T_j^n + \frac{\alpha}{2} T_{j+1}^n.$$

(a)

Again assume

$$T_j^n = \xi^n e^{ikj\Delta x}.$$

Substitution into the Crank–Nicholson equation gives

$$\xi - 1 = \frac{\alpha}{2} \left[e^{ik\Delta x} - 2 + e^{-ik\Delta x} + \xi \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right) \right].$$

Factoring gives

$$\xi - 1 = \frac{\alpha}{2} (1 + \xi) \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right).$$

Using

$$e^{ik\Delta x} - 2 + e^{-ik\Delta x} = -4 \sin^2 \left(\frac{k\Delta x}{2} \right),$$

we obtain

$$\xi - 1 = -2\alpha(1 + \xi) \sin^2 \left(\frac{k\Delta x}{2} \right).$$

Let

$$s^2 = \sin^2 \left(\frac{k\Delta x}{2} \right).$$

Then

$$\xi - 1 = -2\alpha s^2 - 2\alpha s^2 \xi,$$

so

$$\xi(1 + 2\alpha s^2) = 1 - 2\alpha s^2.$$

Therefore,

$$\boxed{\xi(k) = \frac{1 - 2\frac{\kappa\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right)}{1 + 2\frac{\kappa\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right)}}.$$

(b)

Define

$$q = 2\alpha \sin^2 \left(\frac{k\Delta x}{2} \right),$$

where $q \geq 0$. The amplification factor becomes

$$\xi = \frac{1 - q}{1 + q}.$$

Since

$$|1 - q| \leq 1 + q$$

for every $q \geq 0$,

$$|\xi| \leq 1$$

for every wavenumber and every positive timestep.

Thus, like the fully implicit method, the Crank–Nicholson method is unconditionally stable,

Δt is unrestricted by stability.

For sufficiently large timesteps, ξ can become negative for the highest-frequency modes, so those modes may alternate in sign between timesteps. Their magnitude still remains less than or equal to one, however, so this does not violate the stability condition.

4 Problem 4

I implemented the FTCS, fully implicit, and Crank–Nicholson schemes and compared them against the analytic solution for a rod initially at uniform temperature

$$T(x, 0) = 1$$

with both ends held at

$$T(x_{\min}, t) = T(x_{\max}, t) = 0.$$

The numerical grid uses $J + 1$ spatial support points, with the two walls included at $j = 0$ and $j = J$. For all three methods I used

$$\alpha = \frac{\kappa \Delta t}{(\Delta x)^2},$$

with the default conductivity

$$\kappa = 1.$$

For the FTCS method, the update is

$$T_j^{n+1} = T_j^n + \alpha (T_{j+1}^n - 2T_j^n + T_{j-1}^n).$$

For the fully implicit method, the updated temperatures satisfy

$$-\alpha T_{j-1}^{n+1} + (1 + 2\alpha)T_j^{n+1} - \alpha T_{j+1}^{n+1} = T_j^n.$$

For Crank–Nicholson, averaging the explicit and implicit spatial derivatives gives

$$-\frac{\alpha}{2}T_{j-1}^{n+1} + (1 + \alpha)T_j^{n+1} - \frac{\alpha}{2}T_{j+1}^{n+1} = \frac{\alpha}{2}T_{j-1}^n + (1 - \alpha)T_j^n + \frac{\alpha}{2}T_{j+1}^n.$$

The implicit and Crank–Nicholson systems were solved as tridiagonal linear systems for the fixed-boundary case. The implementation also included periodic boundary conditions for the provided periodic test problem.

For the analytic comparison, the solution for a rod of length

$$L = x_{\max} - x_{\min}$$

is

$$T(x, t) = \sum_{\substack{m=1 \\ m \text{ odd}}}^{\infty} \frac{4}{m\pi} \sin\left(\frac{m\pi(x - x_{\min})}{L}\right) \exp\left[-\kappa\left(\frac{m\pi}{L}\right)^2 t\right].$$

I evaluated this series numerically using the first 100 odd modes. The code then compared the numerical solutions to the analytic solution using the RMS difference at each timestep.

$$J = 20$$

For

$$J = 20,$$

the spatial spacing is

$$\Delta x = \frac{1}{20} = 0.05.$$

From the FTCS stability condition derived in Problem 1,

$$\Delta t \leq \frac{(\Delta x)^2}{2\kappa},$$

the largest stable timestep is

$$\Delta t = \frac{0.05^2}{2} = 1.25 \times 10^{-3}.$$

This gives

$$\alpha = \frac{1.25 \times 10^{-3}}{0.05^2} = 0.5.$$

To integrate from $t = 0$ to $t = 1$, this requires

$$\boxed{800 \text{ timesteps}}.$$

All three numerical methods remained stable. The mean RMS differences from the analytic solution over the full run were

$$\begin{aligned} \text{FTCS} &= 1.03388 \times 10^{-3}, \\ \text{Implicit} &= 8.45687 \times 10^{-4}, \\ \text{Crank-Nicholson} &= 1.57402 \times 10^{-4}. \end{aligned}$$

Crank-Nicholson gave the smallest error of the three methods for this test.

$J = 40$ with the same timestep

I then doubled the spatial resolution to

$$J = 40$$

while keeping the previous timestep,

$$\Delta t = 1.25 \times 10^{-3}.$$

The new spatial spacing is

$$\Delta x = \frac{1}{40} = 0.025.$$

Therefore,

$$\alpha = \frac{1.25 \times 10^{-3}}{0.025^2} = 2.$$

This violates the FTCS stability condition

$$\alpha \leq \frac{1}{2}.$$

As expected, the FTCS solution became unstable and eventually overflowed numerically, producing an undefined RMS difference. The fully implicit and Crank–Nicholson schemes remained stable because both are unconditionally stable.

Their mean RMS differences were

$$\begin{aligned}\text{Implicit} &= 8.20621 \times 10^{-4}, \\ \text{Crank–Nicholson} &= 1.04696 \times 10^{-4}.\end{aligned}$$

The numerical overflow in the FTCS solution is consistent with the stability analysis from Problem 1.

$J = 40$ with a stable FTCS timestep

For $J = 40$, the largest stable FTCS timestep is

$$\Delta t = \frac{0.025^2}{2} = 3.125 \times 10^{-4}.$$

This is a factor of four smaller than the stable timestep for $J = 20$, as expected because

$$\Delta t_{\max} \propto (\Delta x)^2.$$

Using this timestep gives

$$\alpha = 0.5$$

and requires

3200 timesteps

to integrate to $t = 1$.

With this timestep, all three methods remained stable. The mean RMS differences from the analytic solution were

$$\begin{aligned}\text{FTCS} &= 2.80008 \times 10^{-4}, \\ \text{Implicit} &= 2.31713 \times 10^{-4}, \\ \text{Crank-Nicholson} &= 4.30775 \times 10^{-5}.\end{aligned}$$

Doubling the spatial resolution therefore required a fourfold reduction in the FTCS timestep in order to maintain stability. The fully implicit and Crank–Nicholson schemes did not require this restriction, and Crank–Nicholson produced the smallest RMS error in each stable comparison.

See Problem4.ipynb or Problem4.pdf.